



VIDYAVARDHINI'S COLLEGE OF ENGINEERING AND TECHNOLOGY
(NBA & NAAC Accredited)
(AUTONOMOUS INSTITUTE AFFILIATED TO UNIVERSITY OF MUMBAI)



VNPS'26

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Title : **Predictive Degradation and Failure
Detection Intelligence System**

Track No :02

Group No: 15

Group Member: Sanchit Gaikwad
Aditya Ghanekar
Omkar Hmabire
Mehul Hingu



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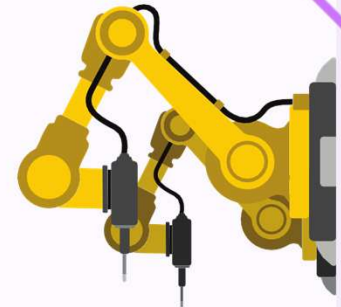


INTRODUCTION

Machines in industries and workshops often fail unexpectedly, leading to production losses, increased repair costs, and safety hazards. Traditional approaches—either reactive maintenance (waiting for failure) or scheduled maintenance—are inefficient and unable to predict faults in advance.

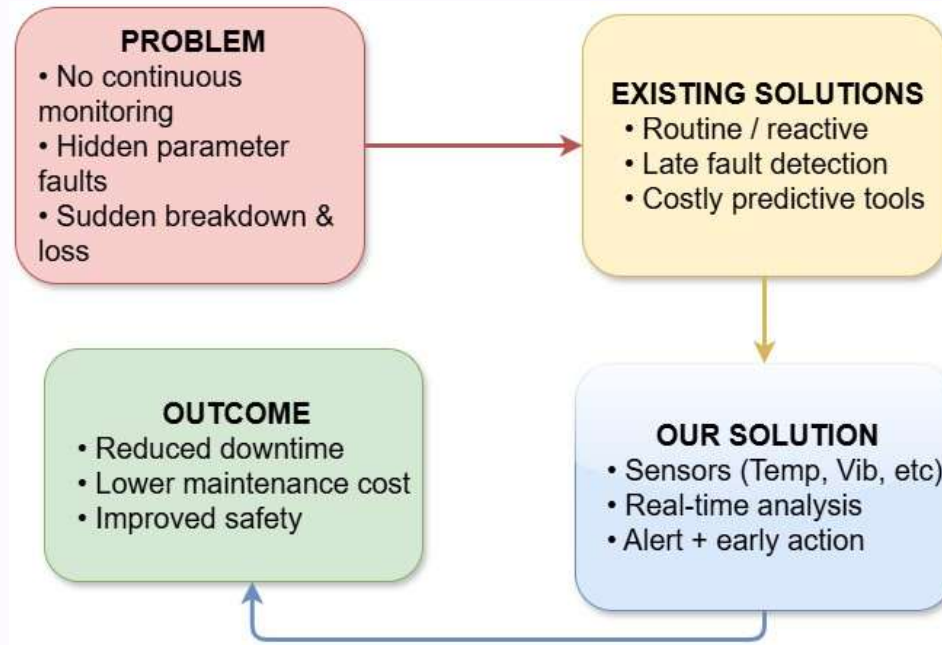
Existing monitoring systems used in large industries are expensive and require skilled operators, making them unsuitable for small industries, workshops, and college labs.

To solve this, we developed a low-cost, portable Machine Health Monitor using Arduino Nano and sensors to detect abnormal conditions early and alert users, preventing breakdowns and improving safety.





ABSTRACT



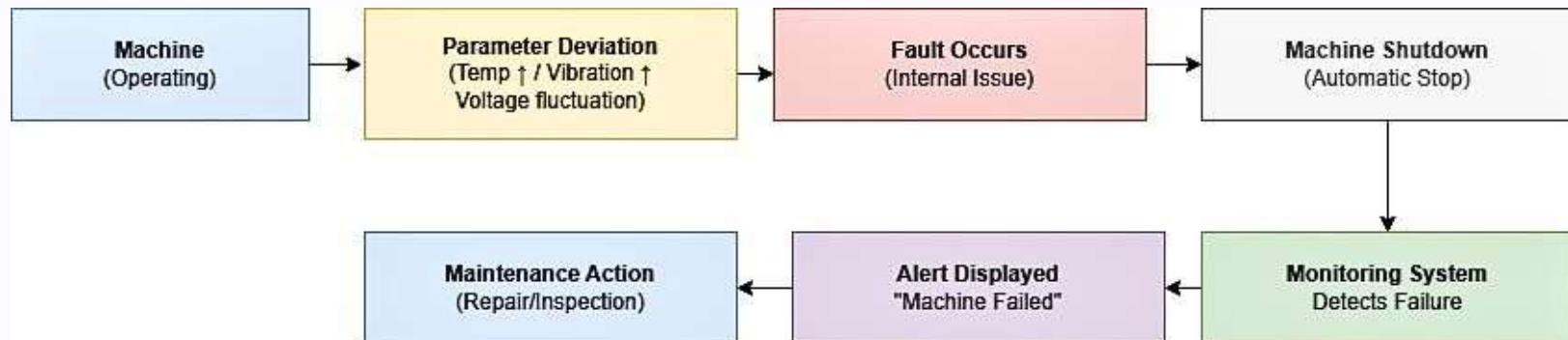


LITERATURE SURVEY

- IoT-based vibration monitoring systems (2019) enable early fault detection but lack multi-sensor integration.
- Machine learning-based predictive maintenance (2020) provides high accuracy but requires large datasets and high computation.
- IoT motor monitoring systems (2021) use temperature and current sensors but miss vibration and sound analysis.
- Acoustic-based fault detection (2018) detects early faults but is sensitive to noise and lacks reliability.

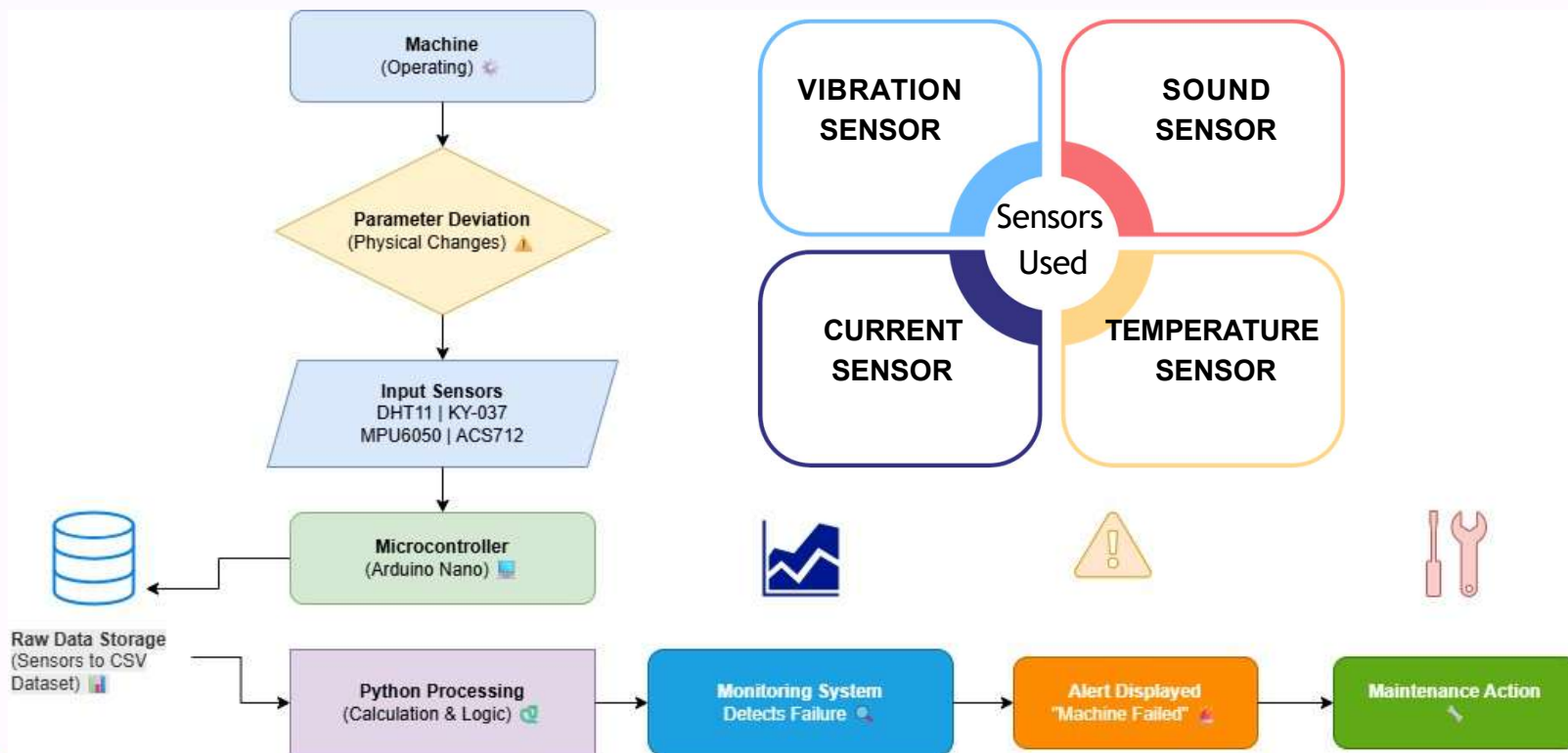


PROBLEM DEFINITION



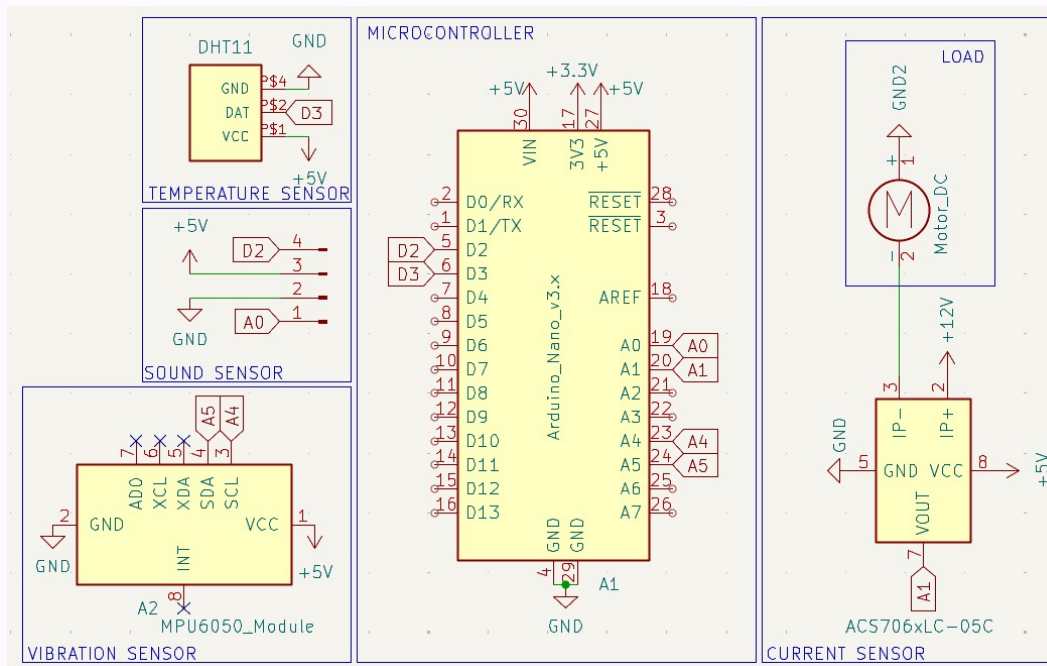


METHODOLOGY





CIRCUIT DIAGRAM

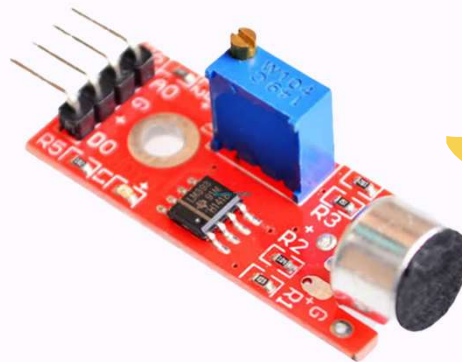




COMPONENTS

1. ACS712 (Current Sensor)

- Range: Typically available in 5A, 20A, or 30A models.
- What Varies: Output Voltage (varies linearly with the sensed AC or DC current).
- Signal Type: Analog.
- COST-(90-110₹)



2.KY-037 High Sensitivity Sound Sensor

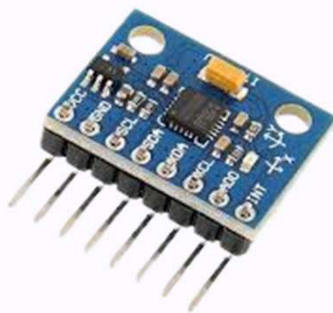
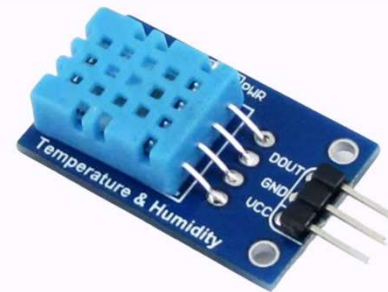
- Range: Adjustable sensitivity via potentiometer; detects roughly 40dB to 100dB.
- What Varies: Output Voltage (for analog) and a High/Low State (for digital threshold).
- Signal Type: Both (has an Analog AO pin and a Digital DO pin)
- COST-(80-100₹).



COMPONENTS

3. DHT-11 Temperature & Humidity Sensor

- Range: 0–50°C (temp) and 20–80% (humidity).
- What Varies: Internal Resistive changes are converted to data by an onboard IC.
- Signal Type: Digital
- COST-(50-70₹)



4. MPU6050 Accelerometer & Gyroscope

- Range: Accelerometer(2g-16g)and Gyroscope (250degree/s to 2000degree/s)
- What Varies: Internal Capacitance changes converted to raw data by an internal 16-bit ADC.
- Signal Type: Digital (I2C Communication).
- COST-(150-170₹)



Industrial Sensor Alternatives

1. Current Sensor (ACS712):

Industrial Alternative:
Hall-Effect Transducers
(e.g., LEM LTS Series)
or TI TMCS1100.

Why: ACS712 has high noise and drift. Industrial versions offer galvanic isolation, higher accuracy (0.1%), and zero temperature drift.

2. Sound Sensor (KY-037):

Industrial Alternative:
Acoustic Emission (AE)
Sensors or Industrial MEMS
Microphones (e.g., Knowles).

Why: KY-037 is too sensitive to ambient noise. AE sensors detect high-frequency friction (ultrasonic) that signals bearing failure long before it's audible.

3. Temp/Humidity (DHT-11):

Industrial Alternative:
PT100/PT1000 RTD (Temp)
and Honeywell HumidIcon
(Humidity).

Why: DHT-11 is slow and inaccurate. RTDs are the industry standard for precision, and can withstand high motor heat.

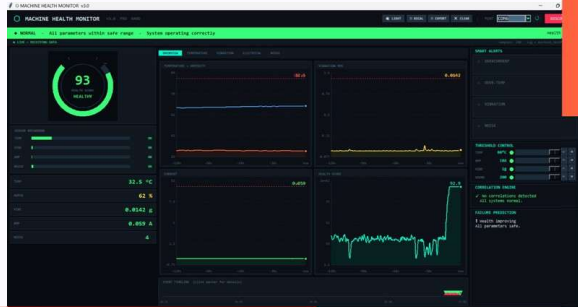
4. Vibration/Motion (MPU6050)

Industrial Alternative:
Piezoelectric Accelerometers
(e.g., ADXL1002 or Fluke 3563).

Why: MPU6050's range and frequency (0-1kHz) are too low. Piezo sensors capture high frequencies (up to 20kHz) needed to detect early-stage bearing pits.

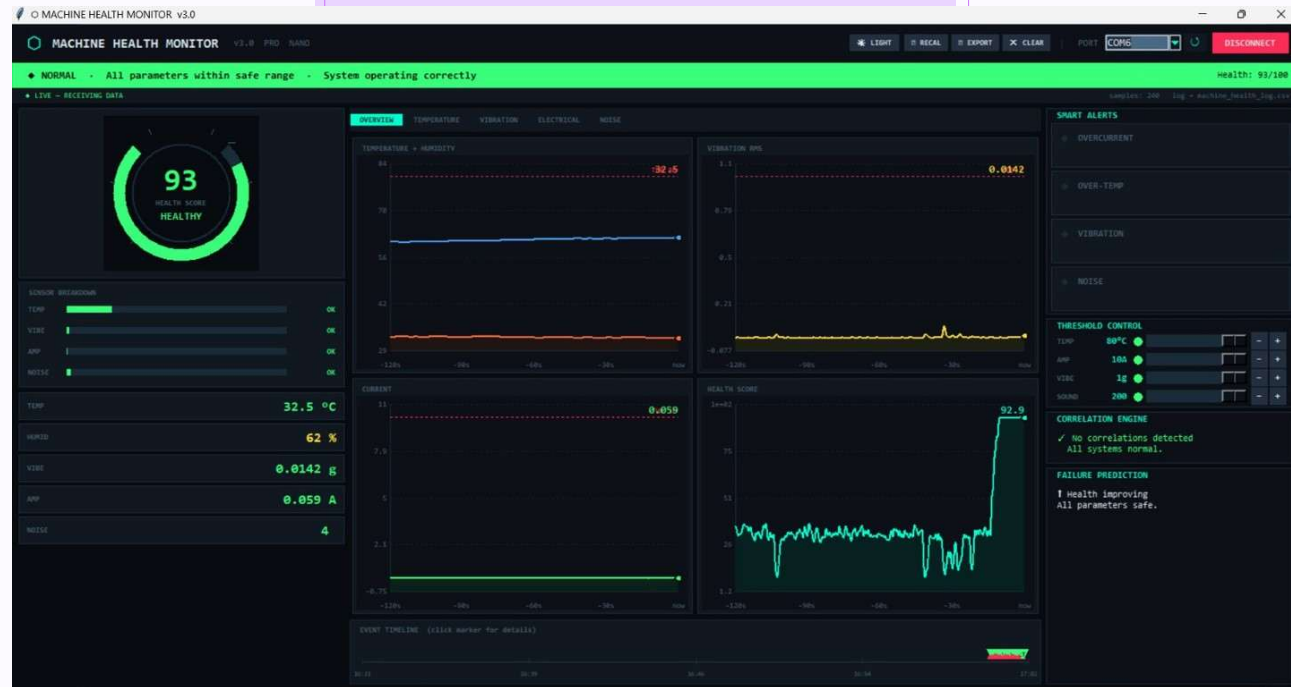


RESULT AND DISCUSSION



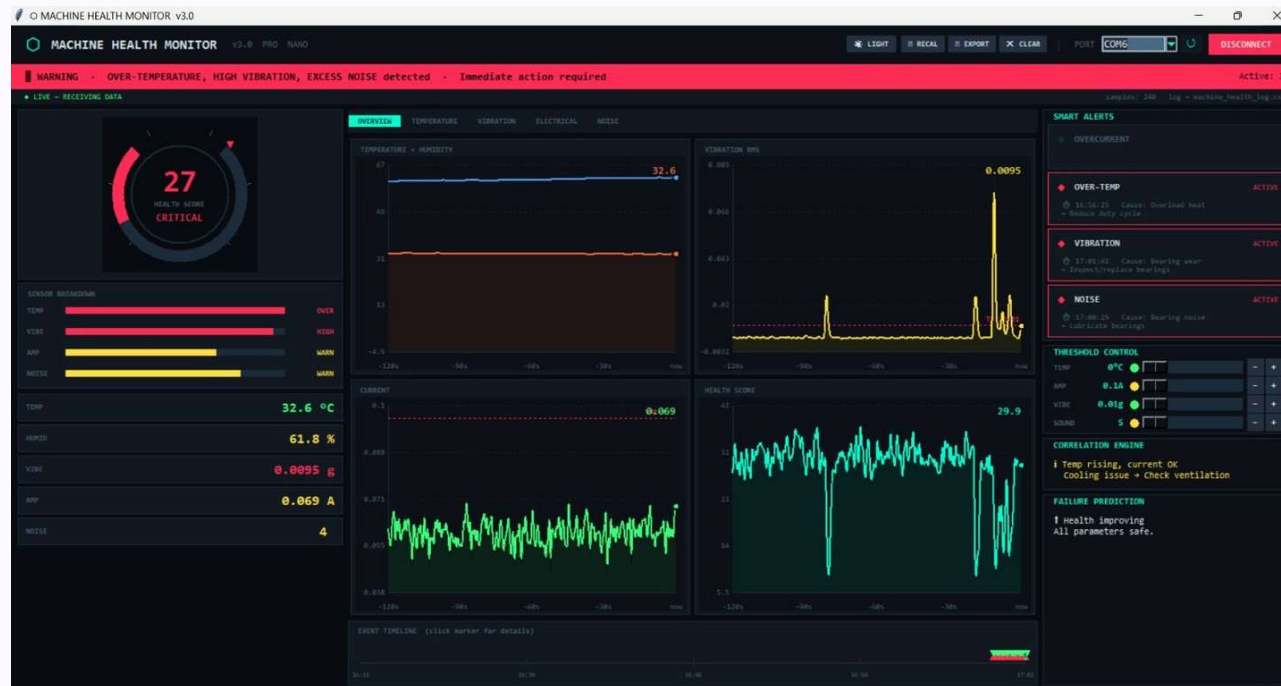


RESULT AND DISCUSSION





RESULT AND DISCUSSION





CONCLUSION

Hence, we Conclude that A real-time machine health monitoring and failure prediction system using multi-sensor data fusion. By integrating temperature, vibration, current, and sound sensing, the system continuously evaluates machine condition. A weighted health score and linear regression-based prediction model enable early detection of faults such as overheating, excessive vibration, electrical instability, and acoustic anomalies. The solution is cost-effective, scalable, and suitable for small-scale industries and academic research, providing a solid foundation for predictive maintenance that can be further improved with advanced signal processing and machine learning.

Integration of advanced machine learning models for improved prediction accuracy

IoT-based cloud connectivity for remote monitoring and data analytics

Future

Scope

Use of advanced signal processing (FFT, wavelet) for precise fault detection

Addition of automated alert and control systems for autonomous operation



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THANK YOU!



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